BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Hexadecimal

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Why does 11111111 plus 00000001 overflow eight bits?

2. A has code 65. Capital-letter codes go in order. What is D's code?

A) 65 B) 68
C) 67 D) 69

3. "Unicode supports fewer characters than standard ASCII." Correct this. Explain why.

4. A program calculates an area and shows a menu in several places. Correct both claims. Explain why. (a) A function cannot return a value. (b) A procedure must return a value.

Today's objectives

Recall the essential ideas of hexadecimal. Apply a clearly modelled method.

Key vocabulary - Read, cover, recall

Hexadecimal

Base-16 representation using 0-9 and A-F.

Nibble

A group of four bits.

Most significant digit

The digit with the highest positional weight.

Denary

Base-10 representation.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Hex uses base 16	Hexadecimal digits are 0-9 and A-F, where A=10 through F=15.
Four bits per digit	One hexadecimal digit represents exactly four binary bits (one nibble).
Binary to hexadecimal	Split an eight-bit value into two four-bit groups and convert each group to one hex digit.
Hexadecimal to denary	Multiply each hex digit by its positional value, 16 and 1, then add the results.

I Do - One worked example

Convert 10101111 from binary into hexadecimal and then denary. [5 marks]

1. Group four bits

2. 1010 = 10 = A and 1111 = 15 = F.

3. Write AF in hexadecimal.

4. In denary, A x 16+F = 10 x 16+15 = 175. [5]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Convert binary 11001010 to hexadecimal. [3]

2. Convert 3E hexadecimal into binary. [3]

3. Convert 2F hexadecimal to denary. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the denary calculation for hexadecimal 2B.

```
first = 2
second = 11
value = ...
print(value)
```

BLACK PEN: Predict or write code; test, then improve.

Question data - complete the working

HEXADECIMAL - split binary into two 4-bit nibbles

1001	0110	Hex: __ __

You Do - Independent exam questions

Q2. Convert binary 10010110 to hexadecimal. [3]

Q3. Convert hexadecimal B4 into eight-bit binary. [3]

Q4. Convert hexadecimal 7D to denary. [4]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A trace table can help check how values change during a program. ☐ T ☐ F
2. Hexadecimal digits are 0-9 and A-F, where A=10 through F=15. ☐ T ☐ F
3. One hexadecimal digit represents exactly four binary bits (one nibble). ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 The true sum is 256, requiring nine bits; only eight stored bits would contain 00000000. ☐

Q2 B - 68 ☐

Q3 Unicode supports far more characters than standard ASCII. ☐

Q4 (a) A function returns a value, such as an area. (b) A procedure performs a task, such as showing a menu, without returning a value. Use arguments and return values to avoid unwanted changes to global data. ☐

Code First - Model

```
first = 2
second = 11
value = first * 16 + second
print(value)
```

Output: 43

We Do - Check the model

1. 1100 = C [1], 1010 = A [1], so CA [1]. ☐

2. 3 = 0011 [1] and E = 1110 [1], so 00111110 [1]. ☐

3. 2 x 16 [1] + 15 x 1 [1] = 32+15 [1] = 47 [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 1001=9 [1], 0110=6 [1], combined hex 96 [1]. ☐

Q3 B=1011 [1], 4=0100 [1], so 10110100 [1]. ☐

Q4 7 x 16 [1] +13 [1] =112+13 [1] =125 [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: